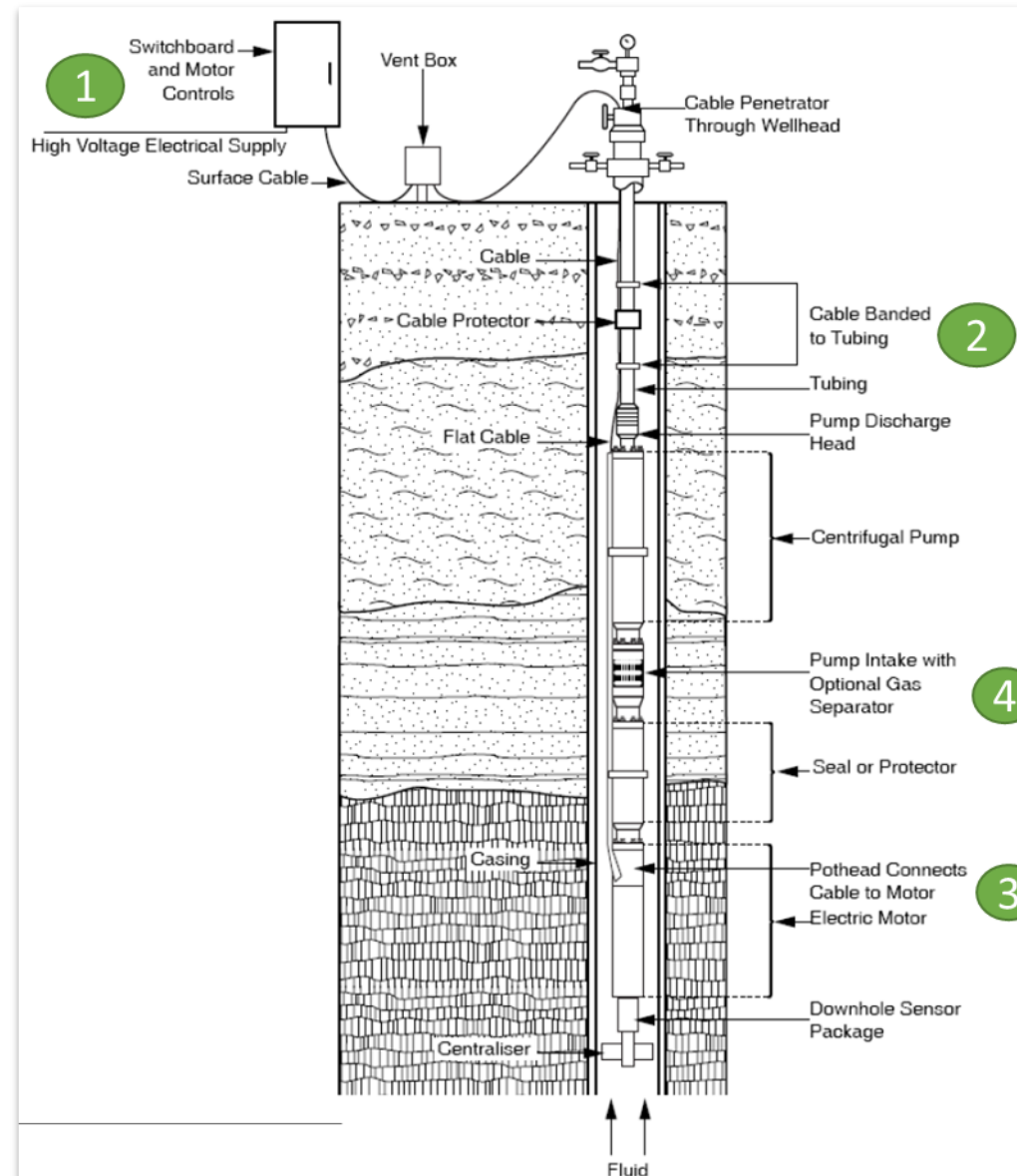




Electrical Submersible pumps (ESP) Failure mechanisms

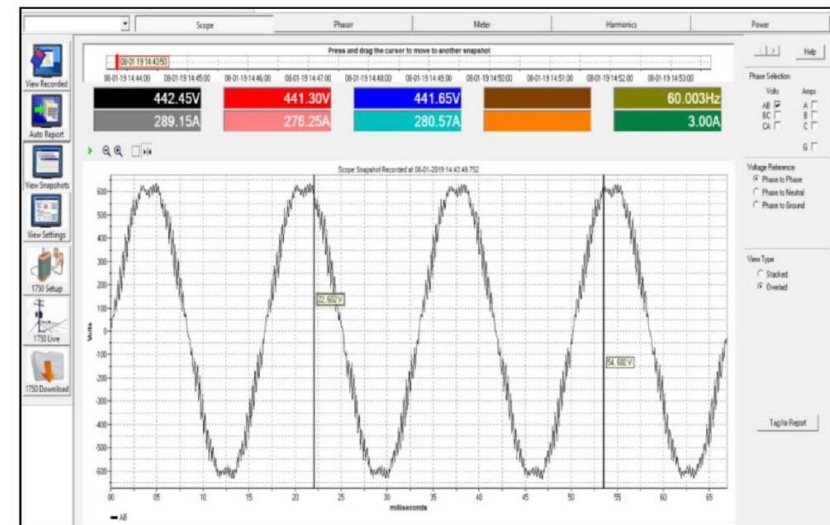
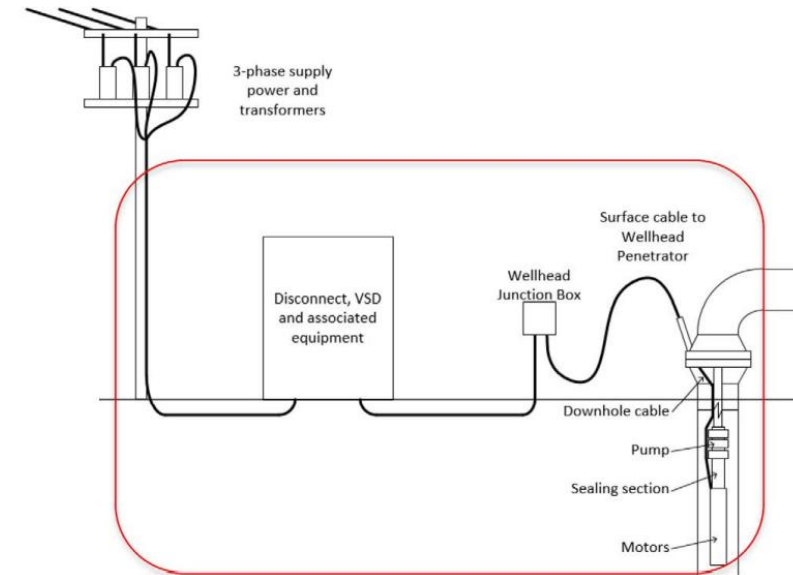


ESP Components



(1) Variable Speed Drive (VSD) & Motor Controls

- It controls the electric output (Frequency, Voltage , Amperage) that allows the ESP to adapt to flow conditions with optimum performance. It also allows soft start of ESP pump and prevent electric surges.
- **Problems**
- If not properly maintained, with a proper harmonic filters , high frequency harmonics can occur at the system causing failures like Penetrator failures, Cable failures, Splice failures / Pothead failures; Motor end turn failures; Motor bearing failures; Loss of sensor readings and; Broken shafts.
- **Mitigation**
- Periodic maintenance and checks for Power quality , installation for appropriate harmonic filters, and common mode filters.



Example of Electric current with high THD content. In this example, it has 5% current THD and 2.7% voltage THD. This is significant because high frequency content on high harmonics are particularly damaging

Power& Monitor Cables

- It delivers the power & Control required to run the pump's motor from surface to downhole.
- **Problems**
- It is a Extremely long cable lengths which increases voltage drop and increases the potential for current reflections (and amplification) , Small diameter conductors, limited insulation (5kV), Multiple connections (splices and pothead) , Use of flat cables resulting in current and voltage imbalance between phases.
- **Mitigation**
- Running & Splicing procedures, slow RIH speeds
- installation of Cable protectors & cross coupling.
- Electric Quality Control with THD (Total harmonic distortion) monitoring



Insulation at cable ends found subjected to scratches/ minor cutting.



Metallic Thread Failure and Electric Cable and Insulation Degradation Corrosion



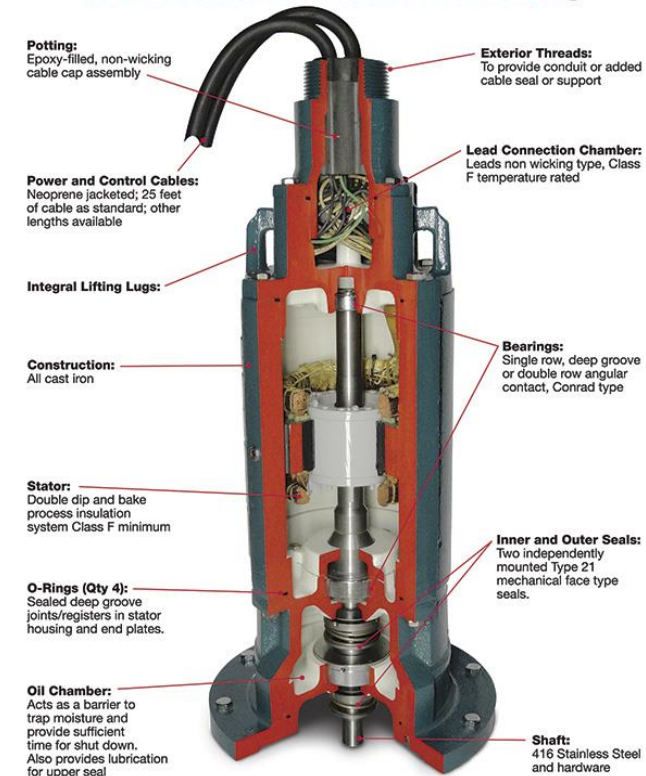
Pothead failure due to common mode current at a reflection point

Motor & Bearings

- Is the part that converts electric energy to a mechanical drive to the pump. In ESP, the motor and the pump are connected by a series of connected Shafts.
- **Problems**
 - This is a delicate part of the pump that is affected by poor power quality (Harmonic control, Ring waves, phase unbalance, current mode currents). Electric shorts, overheating, bearing wear, copper debris accumulation, loss or contamination of dielectric oil.
 - Wrong Flow conditions due to Excessive gases into the pump, accidental closure of valves, tubing holes, Scales that causes pump jamming & motor overheating
 - Corrosion of the casing / threads that is not suitable at elevated temperatures
- **Mitigation**
 - Appropriate Power supply & good quality and continuous monitoring can help the motor run life to extend.

References: SPE-166891-MS, SPE-139810-MS, SPE-194392-MS, SPE-194411-MS

Submersible Motor Cutaway



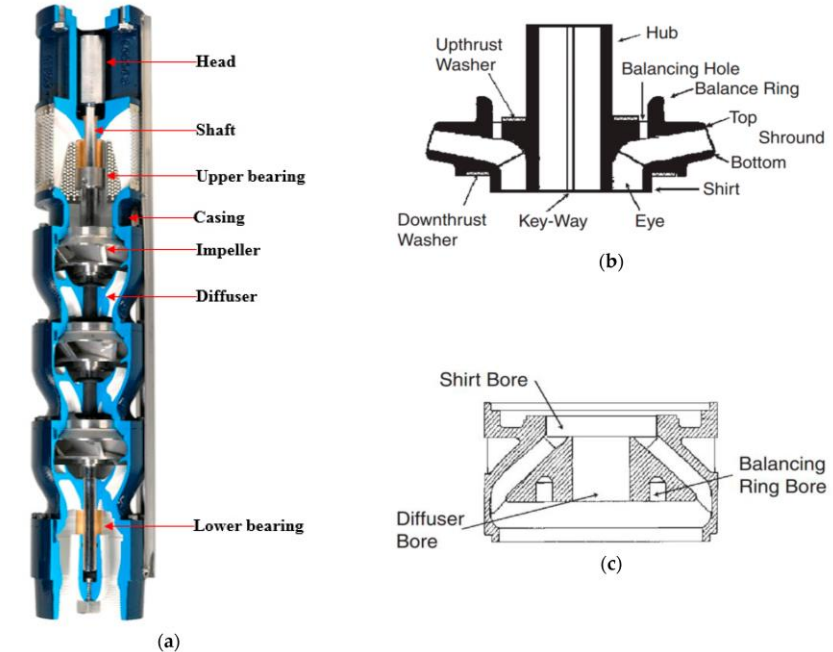
Submersible motor components
(pumpsandsystems.com)



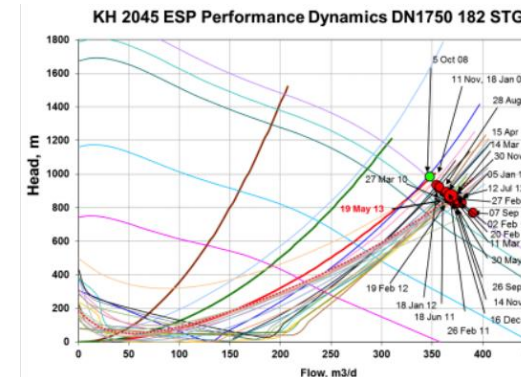
Motor/Bearing Overheating marks. This is proven by loss of Motor insulation. Megger test would indicate Phase to Phase unbalanced or phase to ground is zero.

Pump

- The main part that provides the required pressure to push the fluid to surface
- **Problems**
- Out of range flowing conditions causing upthrust or underthrust of the pump
- Scale plugging is a common cause of premature failures.
- Wrong Flow conditions due to Excessive gases into the pump, accidental closure of valves, tubing holes, Scales that causes pump jamming & motor overheating
- **Mitigation**
- Proper selection of pump size with reasonable operating range along the assumed ESP life
- Downhole injection / for scale inhibitor injection to mitigate scale



Main components of an electrical submersible pump (ESP). (a) Multistage assembly (<http://www.franklinwater.com>); (b) impeller components; (c) diffuser components.



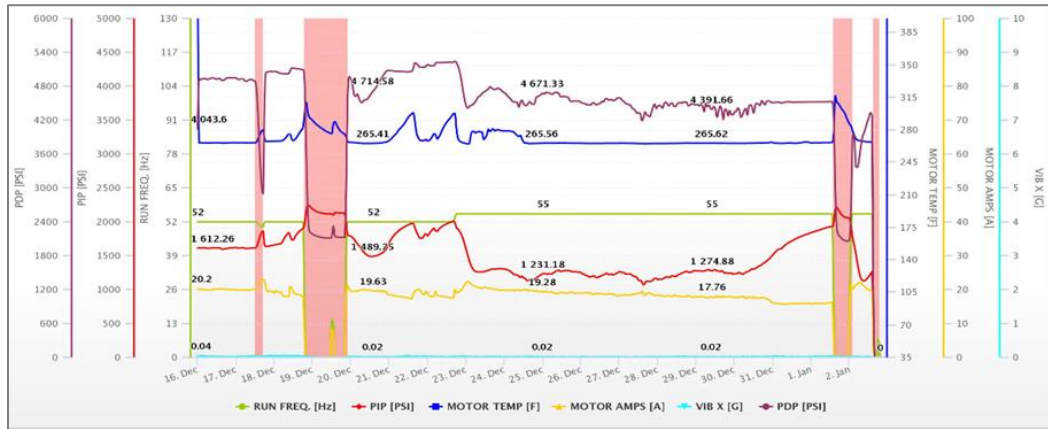
Monitoring systems ensure proactive and fix problems before pump failures



Scale deposition on pump stage shows that it is dominated by Calcium Carbonates

ESP troubleshooting from sensor data

- Effective Monitoring enables the operator to take pro actions before complete failure of ESP (Fixing Electric issues, acid wash,...)



Example of scale deposition and periodic acid wash to prevent ESP failure

Table 1: ESP Operational Problems Identification Checklists

Events	PIP	PDP	ΔP_{pump}	T _m	WHP	Q	AMP	Degradation factor
Blockage at Perforations	↘	↘	↑	↑	↘	↘	↘	↔
Decrease in reservoir pressure	↘	↘	↘	↗	↘	↘	↘	↔
Broken shaft	↑	↓	↓	↗	↓	↓	↓	↑
Wearing Stages	↗	↗	↑	↗	↘	↘	↘	↗
Hole in Tubing	↗	↓	↓	↗	↘	↓	↔	↑
Blockage in Pump Stages	↗	↘	↘	↗	↘	↘	↘	↗
Increase in Reservoir Pressure	↗	↗	↘	↔	↗	↗	↗	↔
Shut in at Surface	↑	↑	↑	↑	↑	↓	↓	↔
Increase in Water Cut	↗	↗	↗	↔	↘	↘	↗	↔
Gas locking	↗	↘	↘	↗	↘	↓	↓	↑

The arrows in Table (1) indicate the rate of change of each parameter while the red arrows indicate the most unique parameter behavior for each case.



Thank you

This is part of course

“Electric Submersible Pumps - Engineering , Operations & Troubleshooting”

If interested, Please email us: info@psa-experts.com